

Accessibility and Inclusive Design Plan

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Status: Draft for accessibility approval

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1. Purpose

Ensure that the product remains usable for readers with limited literacy, low vision, hearing differences, motor limitations, cognitive differences, older devices, and users who prefer simple interaction.

2. Standard

Target WCAG 2.2 AA as the engineering baseline. WCAG emphasizes text alternatives for non-text content and the ability to transform information into speech, symbols, or simpler language, which directly supports this product's audience. [web:30]

3. Inclusive design principles

- Make the linear reader complete.
- Do not require animation.
- Do not require audio.
- Do not communicate by colour alone.
- Use plain language.
- Provide multiple ways to navigate.
- Keep interaction predictable.
- Make the next action clear.
- Avoid shame and stigma.
- Test with real users.

4. Perceivable requirements

Text

- Readable font and line height.
- User-controlled text size.
- No essential text embedded only in images.

- Correct language metadata.
- Sufficient contrast.

Visuals

Every meaningful visual requires:

- Short label.
- Full description.
- Structured target descriptions where relevant.
- Static fallback.
- Text equivalent.

Audio

- Captions or transcript.
- Play/pause.
- Replay.
- Speed control.
- No autoplay by default.
- Audio failure fallback.

5. Operable requirements

- Keyboard access for all controls.
- Visible focus indicator.
- Skip link.
- Logical focus order.
- No swipe-only action.
- No precise corner gesture requirement.
- Pause/stop controls.
- Adequate touch target size.
- No essential time limit.

6. Understandable requirements

- Consistent navigation.
- Clear page and chapter context.
- One idea per screen.
- Plain error messages.
- Predictable language switching.

- Explanation before interaction.
- Clear action labels.

7. Robust requirements

- Semantic HTML.
- Valid heading hierarchy.
- ARIA only when needed.
- Stable accessible names.
- Tested DOM content outside canvas.
- Fallback for unsupported rendering.

8. Motion strategy

Full motion

- Page turns.
- Visual reveals.
- Target focus.
- Optional ambient effects.

Reduced motion

- No page deformation.
- Short opacity or colour transitions only.
- No repeated pulses.
- No parallax.
- Audio highlighting remains understandable.

Lite mode

- Linear reading.
- Static or simple SVG.
- Minimal transitions.
- No WebGL.

Respect the `prefers-reduced-motion` preference and provide an in-app setting. [web:23]
[web:24]

9. Screen-reader behaviour

- Announce publication and lesson title.
- Announce page changes.
- Move focus to the new page heading after navigation.
- Do not announce every decorative animation.
- Expose visual target labels and descriptions.
- Keep transcript readable in DOM.

10. Low-literacy support

- Short sentences.
- Visual sequence.
- Audio narration.
- Replay.
- “Show me” interaction.
- Familiar action words.
- Avoid dense menus.
- Use icons with labels.
- Repeat the safe action.
- Provide facilitator mode.

11. Cognitive accessibility

- Stable navigation placement.
- Limited simultaneous choices.
- Clear progress.
- Avoid surprise motion.
- Explain what happens next.
- Do not hide critical information behind hover.
- Allow pause and return.
- Avoid unnecessary timers.

12. Accessibility testing

Automated

- Axe checks.
- Heading checks.
- Label checks.
- Contrast checks.
- Keyboard focus checks.
- Missing-alt checks.

Manual

- Keyboard-only journey.
- TalkBack.
- VoiceOver.
- Desktop screen reader.
- Browser zoom.
- Text scaling.
- Forced colours.
- Reduced motion.
- Touch target inspection.

User validation

Recruit users with relevant access needs when feasible. Validate whether they can:

- Start a lesson.
- Change language.
- Operate audio.
- Read the visual alternative.
- Navigate linearly.
- Find help.

13. Accessibility acceptance criteria

- All P0 content works in linear mode.
- No P0 keyboard blocker.
- No P0 screen-reader blocker.
- Every visual has a meaningful alternative.
- Captions/transcripts exist for launch audio.
- Reduced motion is functional.

- Text scaling does not clip content.
- Colour-only information is eliminated.